UNIT 3

Exercise 10: Multi-document AI assistant with question answering

Aim

To generate numerical embeddings for text and identify semantically similar sentences using embedding similarity.

Algorithm

- Start the program.
- Load a pretrained sentence-embedding model.
- Define a collection of text documents.
- Convert the documents into numerical vectors.
- Convert the user's query into an embedding.
- Calculate similarity scores between the query and documents.
- Display the most similar document.
- Stop.

Python Code and Output

```
k = min(3, len(all_chunks))
scores, indices = index_multi.search(q_vector, k)

context_parts = []
sources = []

for idx in indices[0]:
    context_parts.append(all_chunks[idx])
    sources.append(source_names[idx])

context = "\n".join(context_parts)
answer = generate_answer(question, context)

print("\nAnswer:", answer)
print("Source files:", ", ".join(sorted(set(sources))))
```

```
Choose Files  2 files
ai_notes.txt(text/plain) - 70 bytes, last modified: 10/27/2026 - 100% done
Experiment_4.docx(application/vnd.openxmlformats-officedocument.wordprocessingml.document) - 71
10/27/2026 - 100% done
Saving ai_notes.txt to ai_notes (2).txt
Saving Experiment_4.docx to Experiment_4 (3).docx
Skipped unsupported file: Experiment_4 (3).docx
Files loaded: 2
Chunks indexed: 1

Ask a question (or type exit): exit
Assistant closed.
```

Result

Text embeddings were generated successfully, and semantic similarity search identified the most relevant document.